

महाराष्ट्र वन सेवा मुख्य परीक्षा - 2025

दि. 08 मे. 2026



2025

R24

BOOKLET NO.

110683

**Forest Services**

**Geology**

**Time Allowed : Three Hours**

**Maximum Marks : 200**

**Medium : English**

**Type of Paper : Conventional/Descriptive**

**Question Paper Specific Instructions**

*Please read each of the following instructions carefully before attempting questions :*

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining questions, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

**Note** – Candidates will be allowed to use Scientific (Non-programmable type) calculators.

**P.T.O.**

**SECTION – A**

- Q1. Solve any five out of seven. (5×8=40)**
- Q1. (a)** Describe in brief physical and chemical weathering processes. **8**
- Q1. (b)** Write an account on primary types of faults with neat diagrams. **8**
- Q1. (c)** Give engineering properties of rocks. **8**
- Q1. (d)** Describe the crystallisation process of albite-anorthite system. **8**
- Q1. (e)** Explain the metamorphic grades and zones. **8**
- Q1. (f)** Enumerate the modes of preservation of fossils. **8**
- Q1. (g)** Give distribution of pre-cambrian rocks of India. **8**
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- Q2. (a)** Give detailed account of landforms developed by river. Draw neat labelled sketches. **15**
- Q2. (b)** Give classification of folds. **15**
- Q2. (c)** Define remote sensing. Describe different photo interpretation elements with suitable examples. **10**
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- Q3. (a)** Describe the textures and structures of igneous rocks with their petrographic significance. **15**
- Q3. (b)** Write a detailed account on metamorphism of arenaceous, argillaceous and basic igneous rocks. **15**
- Q3. (c)** Explain in brief the geological investigations for selection of dam site. **10**
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- Q4. (a)** Write a detailed account on geological history and evolutionary trends in Cephalopoda. **15**
- Q4. (b)** Discuss in detail the tectonic framework of India. **15**
- Q4. (c)** Give basic concepts of geomorphology. **10**
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**SECTION – B**

- Q5.** Solve **any five** out of seven. **(5×8=40)**
- Q5.** (a) Elucidate the methods of radiogenic dating of age determination of earth. **8**
- Q5.** (b) How does the composition of planets and meteorites signify the origin of universe ? **8**
- Q5.** (c) Explain the crystal defects and their significance with a note on elements of X-ray crystallography. **8**
- Q5.** (d) What are the processes of formation of mineral deposits and controls of ore localisation ? **8**
- Q5.** (e) Enumerate the clastic and non-clastic rocks. Add a note on their classification and depositional environment. **8**
- Q5.** (f) What is morphometric analysis of a drainage basin ? Add a note on groundwater recharge. **8**
- Q5.** (g) Elucidate the legislative measures adopted in India for environment protection. Add a note on carbon credit. **8**
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- Q6.** (a) Elucidate the origin, structure and composition of earth and distribution of elements. Add a note on trace element chemistry and its significance. **15**
- Q6.** (b) What are the optical properties of common rock forming minerals ? Elucidate pleochroism, extinction angle, birefringence, twinning and zoning. Add a note on common minerals of igneous and metamorphic rocks. **15**
- Q6.** (c) Explain the process of conservation and utilisation of mineral resources. Add a note on estimation of ore reserves. **10**
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- Q7.** (a) What are sedimentary structures ? Elucidate the sedimentary structures formed on the top of the sedimentary beds and bottom of a sedimentary bed. Add a note on their significance in sedimentology. **15**
- Q7.** (b) What are water bearing characteristics of the rocks ? How porosity, permeability, hydraulic conductivity and transmissivity helps in movement of subsurface water ? Add a note on classification of aquifers. **15**
- Q7.** (c) Discuss the environmental impact of dumping of mine waste and fly ash, industrial and radio-active waste disposal on surface and sub-surface eco-system. **10**
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- Q8.** (a) Discuss in detail the metallogenic epochs and provinces with reference to copper, gold, iron, lead-zinc, manganese, uranium and thorium. Add a note on their ore and gangue minerals. **15**
- Q8.** (b) Elaborate the environmental impact analysis of the natural hazards due to floods, landslides, coastal erosion, volcanoes and their products. **15**
- Q8.** (c) Discuss the classification of crystals into systems and classes of symmetry. Add a note on projection diagrams that represent crystal symmetry. **10**
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